

Project Proposal:

Automated Detection and Mapping of Agricultural-to-Urban Land Conversion in Punjab Using Satellite Imagery

Submitted to: AIRI Team, Punjab Information Technology Board
(PITB)

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1. Background

Punjab has experienced rapid, largely unmonitored conversion of agricultural land into private housing societies over the past decade. Developments such as DHA and Bahria Town, along with numerous smaller housing schemes, have expanded aggressively across peri-urban districts, often without consistent zoning oversight. Currently, tracking this conversion is done manually and inconsistently by relevant departments, with no centralized, data-driven system to measure the scale or rate of farmland loss across the province.

This proposal outlines a computer vision-based system to automatically detect and quantify agricultural-to-urban land conversion across Punjab using multi-temporal satellite imagery, building on remote sensing and deep learning techniques already applied in prior AIRI Team work.

2. Problem Statement

There is currently no automated, province-wide method to measure how much farmland Punjab has lost to real estate development, or to identify which districts are losing agricultural land fastest. This gap limits evidence-based policy responses to unregulated urban expansion.

3. Objectives

- Detect land-use transitions from agricultural to built-up/urban land across Punjab districts using satellite imagery.

- Quantify the rate and scale of farmland conversion per district over a multi-year period.
- Validate detected conversion zones against known housing society developments.
- Produce an interactive, map-based visualization showing conversion patterns over time.

4. Scope

The initial phase will focus on a subset of high-conversion districts (e.g., Lahore, Sheikhupura, Kasur, Faisalabad, Multan, Rawalpindi), with the pipeline designed to scale to all Punjab districts in a later phase.

5. Methodology

1. Data acquisition: Multi-temporal satellite imagery (Sentinel-2, 10m resolution) sourced via Google Earth Engine, covering multiple time points (e.g., 2016, 2019, 2022, 2025).
2. Feature extraction: Computation of spectral indices — NDVI (vegetation) and NDBI (built-up areas) — to distinguish land cover types.
3. Classification: A machine learning classifier (XGBoost/Random Forest, with potential extension to a U-Net segmentation model) trained to label land as agricultural, built-up, or other.
4. Change detection: Comparison of classified land cover across time steps to identify farmland-to-urban transitions at the pixel level.
5. Aggregation and validation: Results aggregated by district and validated against known housing society boundaries to confirm detection accuracy.

6. Visualization: An interactive dashboard (FastAPI backend, Next.js/Streamlit frontend) displaying district-level conversion rates on a map with a time-based slider.

6. Tools and Technologies

Google Earth Engine, Python (rasterio, geopandas), XGBoost/scikit-learn, U-Net (PyTorch/TensorFlow, phase two), FastAPI, Next.js/Streamlit, Leaflet/Mapbox for map visualization.

7. Relevant Experience

This proposal builds directly on prior work completed during this internship, including an end-to-end YOLOv8-based computer vision pipeline (dataset curation, class imbalance handling, model training, and Streamlit deployment) for a helmet compliance detection system. The same technical foundation — data pipeline design, model training, and deployment — applies directly to this satellite-based detection task.

8. Expected Outcomes

- A validated land conversion detection model with documented accuracy metrics.
- A district-level report quantifying agricultural land loss across the study area.
- A deployed, interactive map-based tool for visualizing conversion trends.
- A reusable pipeline that can be extended to additional districts or other land-use monitoring use cases.

9. Timeline

- **Week 1 — Data Pipeline & Model Development:** Set up automated Sentinel-2 retrieval via Google Earth Engine for selected districts (Lahore, Sheikhpura, Kasur, Faisalabad, Multan, Rawalpindi). Compute NDVI/NDBI spectral indices. Train initial classifier (XGBoost/Random Forest) to label land as agricultural, built-up, or other.
- **Week 2 — Change Detection & Validation:** Run classification across multiple time points (e.g., 2019, 2022, 2025). Perform pixel-level change detection to identify farmland-to-urban transitions. Aggregate results by district and validate against known housing society boundaries.
- **Week 3 — Dashboard, Testing & Documentation:** Build the interactive map-based dashboard (FastAPI backend, Streamlit/Next.js frontend) with district-level conversion visualization. Test end-to-end pipeline, finalize district-level report, and prepare documentation/handover materials.
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10. Conclusion

This project would provide PITB and relevant Punjab government departments with a scalable, automated method to monitor agricultural land loss, supporting more informed urban planning and zoning enforcement decisions. It extends existing AIRI Team computer vision capabilities into a new, high-impact application area with direct public policy relevance.